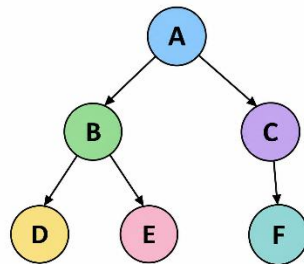


- Q1)** Write a program to implement Depth Limited Search (DLS). **[10 marks]**



- Q2)** Write a program to implement Bernoulli NB classifier with 2 classes on numeric data, use `make_classification` to create a dataset X of 300 sample points in 2D space with the following parameters : **[20 marks]**
`n_features=2, n_informative=2, n_redundant=0, n_classes=2.`
 For testing (Prediction), use the following values to test the Bernoulli NB Classifier.
 Input (Values) : `[[0, 0], [0, 1], [1, 0], [1, 1]]`
 Output : `[0, 1, 0, 1]`

OR

- Q2)** Write a program to implement a Support Vector Machine (SVM) classifier on the Iris dataset. **[20 marks]**
 Input (Dataset and Class Labels):
`X = [ [5.1, 3.5, 1.4, 0.2], [4.9, 3.0, 1.4, 0.2], [4.7, 3.2, 1.3, 0.2], [4.6, 3.1, 1.5, 0.2], [5.0, 3.6, 1.4, 0.2], [7.0, 3.2, 4.7, 1.4], [6.4, 3.2, 4.5, 1.5], [6.9, 3.1, 4.9, 1.5], [6.3, 3.3, 6.0, 2.5], [5.8, 2.7, 5.1, 1.9] ]`
`y = [ 0, 0, 0, 0, 0, 1, 1, 1, 2, 2 ]`
`X_test (Test Data) = [ [5.2, 3.4, 1.5, 0.2], [6.5, 3.0, 4.6, 1.5], [6.2, 3.0, 5.2, 2.0] ]`

- Q3)** Viva **[05 marks]**